

07 Sheet, Task 1.4

Vorgehensweise
für "if and
only if"

$$\|v+w\|^2 = \|v\|^2 + \|w\|^2$$

($\Rightarrow$)

$$\|v+w\|^2 = (v+w)(v+w)$$

assume,
orthogonal

$$= v^2 + \underbrace{2vw}_0 + w^2$$

$$= v^2 + w^2$$

$$= \|v\|^2 + \|w\|^2$$

($\Leftarrow$)

$$\|v+w\|^2 = \|v\|^2 + 2vw + \|w\|^2$$

Assume
equation

$$\Rightarrow \cancel{\|v\|^2 + \|w\|^2} = \cancel{\|v\|^2} + 2vw + \cancel{\|w\|^2}$$

$$2vw = 0$$

$$v \cdot w = 0$$

b)

($\Rightarrow$) Assume orthogonal

$$\begin{aligned} \|u+v+w\|^2 &= (u+v+w)^2 \\ &= (u+v+w)(u+v+w) \\ &= \underbrace{u^2}_{\square} + \underbrace{uv}_{\square} + \underbrace{uw}_{\square} + \underbrace{vu}_{\square} + \underbrace{v^2}_{\square} + \underbrace{vw}_{\square} \\ &\quad + \underbrace{wu}_{\square} + \underbrace{wv}_{\square} + \underbrace{w^2}_{\square} \\ &= u^2 + v^2 + w^2 \\ &= \|u\|^2 + \|v\|^2 + \|w\|^2 \end{aligned}$$

($\Leftarrow$) assume equation

$$\|u+v+w\|^2 = \|u\|^2 + \|v\|^2 + \|w\|^2$$

$$\cancel{u^2 + uv + uw + vu + v^2 + vw} + \cancel{wu + wv + w^2} = \cancel{\|u\|^2 + \|v\|^2 + \|w\|^2} + 0$$

$$uv + vu + uw + wu + vw + wv = 0$$

$$(uv)^2 + (uw)^2 + (vw)^2 = 0$$

As x^2 can't be negative, all
summands must be zero.